

Curvature implies Rockland decay after cancellation

A rigorous one-way answer to the comparison problem in arXiv:2301.08964

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Status: substantial partial theorem, likely valid, pending human review. The curvature assumption implies the full two-sided Rockland estimate after the necessary mean-zero correction. The converse is not proved.

Abstract

Cardona–Delgado–Ruzhansky ask how their Rockland Fourier condition is related to the curvature assumption of Govindan–Sheri–Hickman–Wright. We prove one direction. After subtracting a smooth function with the same mass, the curvature assumption gives both positive- and negative-order Rockland estimates, for the measure and its reflection. The proof isolates an operator-theoretic root lemma and keeps track of the noncommutative order of every alternating convolution. No converse is claimed.

1 Source problem and normalization

Let G be a graded group, let σ be a compactly supported finite measure, and write $\tilde{\sigma}(E) = \overline{\sigma(E^{-1})}$ (omit the complex conjugate for positive measures). The source asks for the relation between the curvature assumption (CA) of [2] and

$$\sup_{\pi \in \widehat{G}} \left\| \pi(\mathcal{R})^{\pm a/\nu} \widehat{\sigma}(\pi) \right\|_{\text{op}} < \infty, \quad (1)$$

where $\mathcal{R}$ is a positive Rockland operator of degree ν .

We use the Fourier convention

$$\widehat{\mu * \eta}(\pi) = \widehat{\eta}(\pi) \widehat{\mu}(\pi), \quad \widehat{\widehat{\mu}}(\pi) = \widehat{\mu}(\pi)^*.$$

Starting with $\sigma^{(0)} = \sigma$, Definition 1.1 of [2] sets

$$\sigma^{(N)} = \begin{cases} \sigma^{(N-1)} * \tilde{\sigma}, & N \text{ odd,} \\ \sigma^{(N-1)} * \sigma, & N \text{ even.} \end{cases}$$

CA says that some $\sigma^{(N)} = h \, dx$ and, for some $\gamma > 0$,

$$\int_G |h(xy^{-1}) - h(x)| \, dx + \int_G |h(y^{-1}x) - h(x)| \, dx \leq C|y|^\gamma. \quad (2)$$

Convolving once more if necessary preserves this estimate and makes N odd. Thus we may take $N = 2m - 1$, in which case the order is exactly

$$\sigma^{(2m-1)} = (\sigma * \tilde{\sigma})^m, \quad \widehat{\sigma^{(2m-1)}} = (A^* A)^m, \quad A = \widehat{\sigma}(\pi). \quad (3)$$

This order matters: in a noncommutative representation it cannot be replaced by $(AA^*)^m$.

Remark 1.8. Now we continue with Remark 1.4. In the Heisenberg group $\mathbb{H}^n$, our results are new in the sense that the criterion in (1.4) allows the use of general Rockland operators $\mathcal{R}$ of higher homogeneous orders. The sub-Laplacian $\mathcal{L} = -\sum_{j=1}^n (X_j^2 + Y_j^2)$ is a Rockland operator of homogeneous order $\nu = 2$. One can consider for instance the positive Rockland operator $\mathcal{R} = \sum_{j=1}^n (X_j^4 + Y_j^4)$. Here $\{X_j, Y_j, Z = \partial_t : 1 \leq j \leq n\}$ denotes the standard basis of left-invariant vector fields of the Lie algebra $\mathfrak{h}_n = \text{Lie}(\mathbb{H}^n)$, see e.g. [22, Chapter 6] for details. It would be interesting to analyse the relation between the condition in (1.5) and other results about the L^p -boundedness of $\mathcal{M}_D^{ds}$ on the Heisenberg group, see [1], [2], [13], [46], [24], [36], [35], and [37]. We also refer the reader to the recent work [26] where a *curvature assumption* was analysed to guarantee the L^p -boundedness of the dyadic maximal function on homogeneous Lie groups.

Remark 1.9. Although the family of graded Lie groups is a sub-class of the family of homogeneous Lie groups (and every homogeneous Lie group is a nilpotent Lie group), every homogeneous Lie group admitting the existence of positive hypoelliptic partial differential operators (also called Rockland operators) is necessarily a graded Lie group in view of the Helffer and Nourrigat solution of the Rockland conjecture, see [28]. So, it is an open problem to determine a relation between the *curvature assumption* in [26] and the group Fourier transform decay property in (1.5) when G is a graded Lie group, or even when the group G is stratified and the Rockland operator $\mathcal{R}$ is considered to be the sub-Laplacian.

Remark 1.10. We observe that even in the Euclidean case, the problem of determining

Figure 1: The comparison problem in Remark 1.9 of [1].

2 Proof intuition

CA turns an alternating convolution power into an L^1 density with a translation Hölder modulus, hence with a small positive Rockland derivative. For an odd iterate its transform is exactly $(A^*A)^m$; a dyadic spectral root argument transfers this derivative first to A^* and, after reflection, to A . Subtracting a smooth function of the same mass preserves those high-frequency estimates, while the resulting cancellation allows the heat-kernel representation of negative Rockland powers. This proves the forward implication only.

3 Three lemmas

Lemma 1 (CA gives a fractional Rockland derivative). *If $h \in L^1(G)$ satisfies (2), then, for every positive Rockland operator $\mathcal{R}$ and every sufficiently small $s > 0$,*

$$\sup_{\pi \in \widehat{G}} \|\pi(\mathcal{R})^{s/\nu} \widehat{h}(\pi)\|_{\text{op}} < \infty.$$

Proof. The translation modulus in (2), integrated against the heat kernel, gives

$$\|(I - e^{-t\mathcal{R}})h\|_1 \lesssim t^{\gamma_0/\nu} \quad (0 < t \leq 1)$$

for some $0 < \gamma_0 \leq \min(\gamma, 1)$. The standard semigroup formula for $\mathcal{R}^{s/\nu}$, split at $t = 1$, is integrable whenever $0 < s < \gamma_0$. Hence $\mathcal{R}^{s/\nu}h \in L^1(G)$. Taking its group Fourier transform proves the assertion. $\square$

Lemma 2 (Root lemma). *Let L be positive self-adjoint on a Hilbert space, let T be bounded with $\|T\| \leq 1$, and suppose*

$$L^s(TT^*)^q \in \mathcal{B}(H)$$

for $s > 0$ and $q \in \mathbb{N}$. Then $L^t T \in \mathcal{B}(H)$ for every $0 < t < s/(2q)$.

Proof. Put $C = TT^*$ and let $E_j = \mathbf{1}_{(2^{-j-1}, 2^{-j}]}(C)$ for $j \geq 0$; the kernel of C makes no contribution to T . Since

$$L^s E_j = L^s C^q (C^{-q} E_j),$$

we have $\|L^s E_j\| \lesssim 2^{jq}$. Interpolation for powers of the positive operator L gives $\|L^t E_j\| \lesssim 2^{jq t/s}$. Also

$$\|E_j T\|^2 = \|T^* E_j T\| = \|E_j T T^* E_j\| \leq 2^{-j}.$$

Consequently

$$\sum_{j \geq 0} \|L^t E_j T\| \lesssim \sum_{j \geq 0} 2^{-j(1/2 - qt/s)} < \infty$$

when $t < s/(2q)$. The spectral sum therefore converges in operator norm to $L^t T$. $\square$

Lemma 3 (Cancellation gives the negative power). *If μ is compactly supported, finite, and $\mu(G) = 0$, then for every positive Rockland operator $\mathcal{R}$ there is $b > 0$ such that*

$$\sup_{\pi \in \widehat{G}} \|\pi(\mathcal{R})^{-b/\nu} \widehat{\mu}(\pi)\|_{\text{op}} < \infty.$$

The same statement holds for $\widetilde{\mu}$.

Proof. Let p_t be the heat kernel of $\mathcal{R}$. At small times $\|\mu * p_t\|_1 \leq \|\mu\|$. At large times, $\mu(G) = 0$, heat-kernel scaling, and a first positive translation modulus give, for some $\delta > 0$,

$$\|\mu * p_t\|_1 \lesssim t^{-\delta/\nu} \quad (t \geq 1).$$

Thus the Bochner integral

$$\mathcal{R}^{-b/\nu} \mu = c_b \int_0^\infty t^{b/\nu-1} \mu * p_t dt$$

converges in L^1 whenever $0 < b < \delta$. Its Fourier transform is the operator in the statement. Reflection gives the last assertion. $\square$

4 The one-way comparison theorem

Theorem 4. *Let σ be a compactly supported finite measure on a graded group G and suppose that σ satisfies CA. Choose $\phi \in C_c^\infty(G)$ with $\int_G \phi = 1$ and put*

$$\mu = \sigma - \sigma(G)\phi.$$

For every positive Rockland operator $\mathcal{R}$ of degree ν there is $a > 0$ such that

$$\begin{aligned} \sup_{\pi \in \widehat{G}} \|\pi(\mathcal{R})^{\pm a/\nu} \widehat{\mu}(\pi)\|_{\text{op}} &< \infty, \\ \sup_{\pi \in \widehat{G}} \|\pi(\mathcal{R})^{\pm a/\nu} \widehat{\widetilde{\mu}}(\pi)\|_{\text{op}} &< \infty. \end{aligned}$$

Thus the mass-corrected measure and its reflection satisfy the Rockland condition in [1].

Proof. Use the odd iterate (3). Lemma 1 gives, uniformly in π ,

$$\pi(\mathcal{R})^{s/\nu} (A^* A)^m \in \mathcal{B}(H_\pi).$$

After a harmless common normalization of A , Lemma 2 with $T = A^*$ and $q = m$ yields a positive Rockland estimate for A^* .

For completeness, CA also passes to $\tilde{\sigma}$. Indeed, $(\tilde{\sigma} * \sigma)^{m+1} = \tilde{\sigma} * (\sigma * \tilde{\sigma})^m * \sigma$; convolution of the density h on the left and right by compactly supported finite measures preserves the two translation estimates in (2). Applying the same argument to $\tilde{\sigma}$, whose transform is A^* , gives the positive Rockland estimate for A .

The correction by ϕ preserves both positive estimates because every positive Rockland power of a compactly supported smooth function is in L^1 . Since $\mu(G) = 0$, Lemma 3 gives the two negative estimates. Take a smaller than the four exponents obtained above. $\square$

5 Scope and obstruction

The correction is not cosmetic. A negative Rockland power at the trivial representation forces zero mass; hence an arbitrary nonzero-mass surface measure cannot satisfy the symmetric condition before cancellation. The smooth maximal term introduced by the correction is controlled by the usual Hardy–Littlewood maximal operator, exactly as in [2].

No converse is proved. Abstractly, a one-sided estimate $L^a A \in \mathcal{B}(H)$ does not control a positive derivative of $A^* A$: take $Le_n = 2^n e_n$ and a rank-one map with smooth range but with adjoint range outside every positive Sobolev domain. Realizing such a field as the group Fourier transform of a compact measure is a separate problem, so this is an operator-theoretic warning rather than a group counterexample.

The result is therefore classified as a substantial partial answer: it proves $\text{CA} \Rightarrow \text{mass-corrected Rockland decay}$, but neither the reverse implication nor the weak $(1, 1)$ endpoint.

References

- [1] F. Cardona, S. Delgado, and M. Ruzhansky, *Boundedness of the dyadic maximal function on graded Lie groups*, arXiv:2301.08964v2 (2024).
- [2] A. Govindan Sheri, J. Hickman, and J. Wright, *Lacunary maximal functions on homogeneous groups*, J. Funct. Anal. 286 (2024), 110250; arXiv:2305.04700.